

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **Python** data pipelines on **AWS Lambda** to process and transform large volumes of advertising event data, handling approximately **3,500 events per second** during peak hours.
- Developed and deployed **scalable Python services** using **FastAPI** for real-time data inspection and bias investigation within multimedia data pipelines, integrating with annotation tools.
- Built **data visualization dashboards** using **Power BI** to present model quality metrics and error patterns to data science and engineering teams, improving data insights accessibility.
- Authored **production runbooks** and contributed to **CI/CD pipelines** for automated deployment and monitoring of ML data processing services on **AWS**.
- Collaborated with **data scientists** and **linguists** to define requirements for data quality checks and ontology services, ensuring alignment with ML research objectives.
- Containerized **Python applications** with **Docker** and managed deployments to a **Kubernetes** cluster (EKS), configuring basic resource limits and monitoring.
- Implemented **robust database designs** and optimizations in **PostgreSQL** for storing and querying large datasets used in ML model training and evaluation.

SKILLS

Languages – Python, SQL, Java, R, English

Cloud & Big Data – AWS (EC2, S3, Lambda, CloudWatch), Databricks, Spark, Hadoop

ML & Data Engineering – Machine Learning, Data Pipelines, ETL, scikit-learn, pandas, NumPy, ML Ops, Data Visualization, Feature Engineering, Model Evaluation

Databases – PostgreSQL, MySQL, Oracle

Tools & Technologies – Docker, Kubernetes, Jenkins, Git, Github, Jira, Confluence, Power BI, Tableau, RStudio, FastAPI, Flask

Methodologies – Agile, Scrum, CI/CD

PROJECTS

ML Feature Pipeline for Ad Inventory Analysis

- Engineered a **Python-based feature pipeline** on **AWS S3** and **Lambda** to extract behavioral signals from ~1.5M advertising impression records.
- Trained a **LightGBM classifier** on the extracted features to identify brand safety risks, achieving approximately **0.91 AUC** and **0.45 F1-score** on a held-out dataset.
- Developed a **FastAPI service** to serve model predictions with input validation and p95 latency under 200ms, supporting downstream analysis of advertising inventory.

Interactive Dashboard for Data Quality Monitoring

- Designed and built interactive **Power BI dashboards** to visualize key data quality metrics and identify bias patterns across multiple advertising data sources.
- Implemented **DAX measures** and **Power Query** transformations to aggregate and clean data from diverse sources, supporting data inspection workflows.
- Collaborated with data scientists to refine dashboard requirements, ensuring clear presentation of data anomalies and model performance indicators to stakeholders.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)